Week 8: Options Market Making & Dynamic Hedging

Avellaneda-Stoikov with a vector-valued (Greek) inventory

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Contents

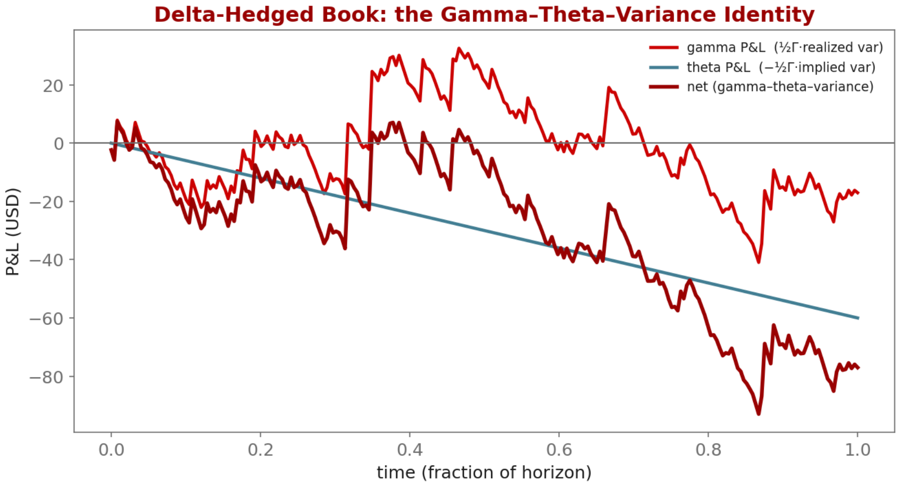
Learning Objectives

- Set up the El Aoud-Abergel multi-Greek framework
- Understand the quadratic inventory penalty
- Quantify the P&L of a delta-hedged book

- Inventory becomes a vector $q = (\text{Delta}, \text{Gamma}, \text{vega}, \dots)$
- Quadratic inventory penalty; reduces to AS in 1-D
- Multi-Greek HJB and the separation ansatz

- Continuous vs discrete delta hedging
- The gamma-theta-variance identity
- Perp futures as the hedge instrument; funding cost

Illustration



$$\text{penalty} = \tfrac{1}{2}\, q^{\mathsf{T}} \Sigma\, q$$

Why: Inventory risk for an options book is multi-dimensional (delta, gamma, vega...). A quadratic form is the local (mean-variance) risk of the Greek vector; Σ is calibrated to the empirical covariance of the Greeks, so cross-risks are priced, not just diagonal ones. It reduces to AS's q^2 penalty when there is one Greek.

$$d\Pi = \tfrac{1}{2}\,\Gamma\, S^2\left(\sigma^2_{\text{real}} - \sigma^2_{\text{impl}}\right) dt$$

Why: The gamma-theta-variance identity: a delta-hedged option earns (or pays) the gap between realized and implied variance, scaled by dollar gamma. This single equation explains 80

Getting the Data from Coincall

- Greeks per contract (for the inventory vector and Sigma):
  - `GET /open/option/detail/v1/{symbol} -> delta, gamma, vega, theta, rho`
- Hedge instrument + carry:
  - `Perp book: GET /open/futures/order/orderbook/v1/{symbol}`
  - `Funding: GET /open/public/fundingRate/v1?symbol=BTCUSD`
- Offline: coincall_funding_rates_2025Q4.parquet for hedge-cost accounting

Cross-check exact paths at https://docs.coincall.com/

Code — week8_hedge.py

```
import torch
def book_greeks(positions, greeks):
    # positions: (n,), greeks: (n,5) columns [delta,gamma,vega,theta,rho]
    return (positions[:, None] * greeks).sum(0) # aggregate inventory vector

def delta_hedge(net_delta, perp_pos, threshold=0.1):
    if net_delta.abs() > threshold: # rebalance only when breached
        trade = -net_delta # hedge with perp futures
        return perp_pos + trade, trade
    return perp_pos, torch.tensor(0.0)
```

Glossary — Key Terms

Greek inventory vector aggregated (delta, gamma, vega, ...) exposure of the book

Quadratic penalty the risk term (1/2) q^T Sigma q on the Greek vector

Delta hedging neutralizing directional risk using the underlying or a perpetual

Gamma scalping P&L earned by re-hedging a long-gamma position

Gamma-theta-variance identity hedged P&L = (1/2) Gamma S^2 (realized var - implied var) dt

Funding rate periodic payment between perpetual-swap longs and shorts

El Aoud-Abergel the multi-Greek option market-making framework

Reading

- El Aoud-Abergel (2015); Taleb (1997); Cartea et al., Ch. 11

References

- El Aoud, S., & Abergel, F. (2015). A stochastic control approach to option market making. Market Microstructure and Liquidity, 1(1), 1550006.
- Taleb, N. N. (1997). Dynamic Hedging: Managing Vanilla and Exotic Options. Wiley.
- Cartea, Jaikumar, & Penalva (2015). Algorithmic and High-Frequency Trading, Ch. 11.